Universidade de Coimbra
Faculdade de Ciências e Tecnologia
Departamento de Engenharia Informática

Knowledge and Language 2025/2026

Masters in Informatics Engineering

Project

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Attention: plagiarism will not be tolerated!

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1 Goals

The main goal of the *Knowledge and Language* project is **to address a challenge** within the scope of the topics covered in the course. The core requirement is that the project must involve both of the following components:

- **Knowledge Representation**
- **Natural Language Processing**

Any computational tools and data sources may be used, provided that their use is well justified. The chosen domain is open and can be freely selected by the students.

2 Stages

The project should be developed by groups of **two students**[1], and must go through the following **stages**:

1. **Research & Problem Identification:** defining the challenge to tackle, supported on scientific literature. Two deliverables are expected, both of which can be later integrated in the **final report**:

   * **Project Proposal:** a single-page document describing the problem to tackle, setting the **goals** of the project. The proposal must further enumerate, clearly, the **data** and **tools** to be used, proposed approach(es), as well as connections with the reviewed literature (e.g., initial inspiration, what will be made different) as well as with the topics covered by the course. Additionaly you may include the **dates of intermediate checkpoints**.
   * **Literature Review:** a short report, with no more than **three pages in a 2-column format**[2], reviewing one or more recent papers on the topics of the course. At least one paper should be less than 5 years old and related to the project. Among others, the review must highlight the connections with the topics covered by the course.

[1] Please indicate the members of your group in the following shared document: `https://tinyurl.com/kl-groups`

[2] Formatted according to the IJCAI guidelines, `https://www.ijcai.org/authors_kit`

2. **Project Execution:** comprising the development of the project (design, implementation, evaluation), always **meeting the checkpoints** set and their discussion with the Professor.

3. **Final Report:** all the work developed must be described in a **final document**, in the form of a scientific paper, with no more than **eight 2-column pages**[3]. The following structure is suggested:

   (a) Introduction (including motivation, problem description, goal and approach; may be adapted from the Project Proposal);

   (b) Related Work (for which the Literature Review can be a starting point);

   (c) Data & Approach (an overview of the data used and of the approach adopted for tackling the problem);

   (d) Implementation (including data exploitation, applied algorithms and tools used);

   (e) Experimentation (examples of good and bad results, quantitative evaluation when possible, and their discussion);

   (f) Conclusions (stressing the main achievements and main difficulties faced) and Future Work (what could still be done);

   (g) Bibliographic References.

4. **Final Presentation:** in the end of the semester, there will be 10-minute (tentative) presentations of the projects developed, followed by a brief discussion between the Professor and colleagues (if present).

All documents can be **written either in Portuguese or English**.

3 Evaluation Criteria

Overall, the project is worth **60%** of the final grade of the *Knowledge and Language* course (**12/20**), split in the following:

- Research & Problem (**2/20**, incl. Literature Review, Project Proposal);
- Checkpoints (**3/20**, **at least two**, assessed in Project Support classes);
- Final (**7/20**, incl. Final Report, Presentation).

Evaluation will focus on, but not be limited to, the following aspects:

- Meeting high-level **requirements**.

[3]Ideally, seven pages for content plus one for bibliographic references.

- Meeting **checkpoints** and **goals** set in the Project Proposal.
- Clarity of the produced **documentation**.
- Proper application of **course-related concepts**.
- **Novelty** of the proposed solution.
- **Complexity** of the developed work.
- **Quality** and **meaningfulness** of results / discussion / conclusions.
- **Final presentation** and **discussion**.

Further remarks:

- **Missing a deadline** will result in a **penalty**. This includes the **submission** of the Literature Review, Project Proposal, Final Report, and **checkpoints**.
- The **Final Presentation is mandatory**.
- Even though the project is to be developed in a group, the final grade may be **different for elements of the same group**, depending on individual performance.
- The work must be authored exclusively by the group, with each member assuming responsibility for the materials presented. The use of writing assistance tools or code reuse must be explicitly stated and carried out with critical awareness. It is mandatory to cite all sources used, including books, articles, web pages, AI Tools, or any other format through which third-party work is incorporated.

4 Deadlines

The project has **two** main (hard) deadlines, with results submitted to *Inforestudante*:

- [**17th October 2025**] Literature Review & Project Proposal
- [**5th December 2025**] Final Report

Two dates must be set by the group in the Project Proposal for checkpoints, where progress will be presented to and discussed with the Professor. These can be selected among the following dates:

- October 21, 28
- November 4, 11, 18, 26

- December 2

Schedule your checkpoint dates at the following **Google Calendar link**[4].

Both members of the group **must** be present in all checkpoints.

Oral presentations should coincide with the final classes, but additional slots may be necessary to accommodate for all groups:

- 9^{th} **December 2025**
- 16^{th} **December 2025**

A Suggested Topics – Starting Points, Not Limits

As long as the main requirements are met, the group is free to choose the topic and goals for their project, which should, nevertheless, be discussed with and validated by the Professor. Here are some generic suggestions of possible topics:

- **Knowledge Graph Construction** [1]
  Build a structured representation of knowledge from raw textual documents by identifying relevant entities and their relations. This knowledge can then support querying, reasoning or downstream applications.
  Example domains: music, tourism, scientific abstracts, etc.
- **Knowledge Graph Enrichment** [2]
  Start from an existing Knowledge Graph and enrich it with new entities and triples extracted from textual documents or LLMs. Evaluate the quality and relevance of the enrichment process.
- **Triple Extraction and Ontology Learning from LLMs** [3]
  Use prompt engineering or fine-tuned models to extract structured facts from natural language. Represent this knowledge in RDF or OWL and validate it with SPARQL queries. Consider participation in challenges like LLMs4OL[5] or LM-KBC[6].
- **Knowledge Injection into LLMs** [4]
  Select a task (e.g., text classification or QA) and analyze the impact of injecting external symbolic knowledge (e.g., triples, ontology concepts) into the LLM's reasoning process.
  Example domains: education, health, climate, legal advice, etc.

[4] `https://calendar.app.google/8QqtvqmrwvVDUWrXA`
[5] `https://sites.google.com/view/llms4ol`
[6] `https://lm-kbc.github.io/challenge2025/`

- **Knowledge Graph Embeddings for NLP Tasks** [5]
  Embed a symbolic knowledge graph using techniques like RDF2Vec or TransE, and use those embeddings in downstream NLP tasks. Compare results using symbolic vs. embedded representations.

- **Conversational Agent over Custom Knowledge** [6]
  Develop a chatbot that can answer questions based on a knowledge base created from a specific set of documents. The project may focus on integrating retrieval mechanisms with generative models or using structured queries.
  Example domains: internal company documentation, study notes, product manuals, etc.

- **Reasoning under Uncertainty or Incomplete Knowledge** [7]
  Explore how to model and reason with data that is noisy, ambiguous or incomplete. This may involve symbolic approaches (e.g., probabilistic logic, fuzzy rules) or the use of LLMs with uncertainty-aware mechanisms.
  Example domains: misinformation detection, medical diagnosis, social media analysis

  Note: Every project should include a small evaluation component to support its conclusions. This may involve the use of a known benchmark dataset, comparison against a baseline system, or the design of a custom evaluation methodology (e.g., human annotation, task-specific metrics).

Students are encouraged to define their own project within the scope of the course, and may freely combine aspects from multiple suggestions and propose a new problem, as long as it involves both core components of the course.

References

[1] Shaoxiong Ji, Shirui Pan, Erik Cambria, Pekka Marttinen, and Philip S Yu. A survey on knowledge graphs: Representation, acquisition, and applications. *IEEE transactions on neural networks and learning systems*, 33(2):494–514, 2021.

[2] Xiaoyan Zhao, Yang Deng, Min Yang, Lingzhi Wang, Rui Zhang, Hong Cheng, Wai Lam, Ying Shen, and Ruifeng Xu. A comprehensive survey on relation extraction: Recent advances and new frontiers. *ACM Computing Surveys*, 56(11):1–39, 2024.

[3] Badr AlKhamissi, Millicent Li, Asli Celikyilmaz, Mona Diab, and Marjan Ghazvininejad. A review on language models as knowledge bases. *arXiv preprint arXiv:2204.06031*, 2022.

[4] Garima Agrawal, Tharindu Kumarage, Zeyad Alghamdi, and Huan Liu. Can knowledge graphs reduce hallucinations in LLMs? : A survey. In Kevin Duh, Helena Gomez, and Steven Bethard, editors, *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3947–3960, Mexico City, Mexico, June 2024. Association for Computational Linguistics.

[5] Xiou Ge, Yun Cheng Wang, Bin Wang, C-C Jay Kuo, et al. Knowledge graph embedding: An overview. *APSIPA Transactions on Signal and Information Processing*, 13(1), 2024.

[6] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474, 2020.

[7] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. *arXiv preprint arXiv:2109.07958*, 2021.